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3.2.3 Economic Performance

Economic Growth: Actual vs Potential

Economic growth is an increase in the real GDP of an economy over time — meaning the economy is producing more goods and services. It is the most important indicator of improving living standards.

Actual growth — an increase in real GDP; the economy moves towards or along its PPF, utilising previously unemployed or underemployed resources. Shown by a movement from inside the PPF to a point on it, or by an increase in AD.

Potential (trend) growth — an increase in the productive capacity of the economy. The PPF shifts outward, or LRAS shifts right. This means the economy *could* produce more, even if it isn't doing so yet.

Feature	Short-Run (Actual) Growth	Long-Run (Potential) Growth
What increases	Real GDP (actual output)	Potential/trend output (productive capacity)
Caused by	Increase in AD (any component)	Increase in LRAS (quality/quantity of factors of production)
Diagram	Movement towards or along the PPF; AD shifts right	Outward shift of the PPF; LRAS shifts right
Inflation risk	May be inflationary if the economy is near full capacity	Non-inflationary (capacity expands alongside output)

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Sustainability	Unsustainable if capacity isn't also growing	Sustainable — permanent increase in capacity

Benefits of Economic Growth

- **Higher living standards:** More goods and services available per person. Real incomes rise, allowing people to consume more and enjoy greater material wellbeing.
- **Lower unemployment:** Growing output requires more workers (derived demand for labour), reducing unemployment and its associated social costs.
- **Higher tax revenue:** Even without raising tax rates, a growing economy generates more income tax, VAT, and corporation tax revenue. This gives the government more resources for public services or debt reduction.
- **Fiscal dividend:** The automatic improvement in government finances during growth (higher tax revenue + lower benefit spending) gives policymakers room to invest or cut taxes.
- **Improved business confidence:** Growth encourages firms to invest, creating a virtuous cycle of rising demand → rising investment → rising output.
- **Poverty reduction:** Even if growth is imperfectly distributed, it tends to lift absolute living standards across the income spectrum.

Costs and Risks of Economic Growth

- **Environmental degradation:** Higher production often means more pollution, carbon emissions, deforestation, and resource depletion. GDP growth that destroys natural capital may reduce future welfare.
- **Inflation:** If AD grows faster than productive capacity (demand-pull inflation), growth causes rising prices that erode purchasing power.
- **Inequality:** Growth may disproportionately benefit the wealthy (capital owners, highly skilled workers), widening the gap between rich and poor. The phrase "a rising tide lifts all boats" doesn't hold if some people don't have boats.
- **Current account deterioration:** Growing incomes suck in imports, potentially worsening the trade balance.
- **Opportunity cost:** Growth driven by investment in capital goods means sacrificing current consumption. Growth driven by debt-fuelled consumption is unsustainable.

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- **Quality of life:** Higher GDP may come at the cost of longer working hours, stress, congestion, and less leisure time — none of which are captured by GDP statistics.

Real-World Example: China's Growth Miracle and Its Costs

China averaged roughly 10% annual GDP growth from 1980 to 2010 — the fastest sustained growth in human history. This lifted over 800 million people out of extreme poverty and transformed the country into the world's second-largest economy. However, the costs were severe: catastrophic air pollution (Beijing smog regularly hit hazardous levels), mass displacement of communities for infrastructure projects, extreme inequality between coastal cities and rural interior regions, and an authoritarian political system that suppressed dissent. China's experience illustrates that growth brings enormous benefits but also enormous costs — the composition and distribution of growth matter as much as its rate.

The Output Gap

The output gap is the difference between actual GDP and potential GDP (the level of output the economy could produce at full employment).

Positive output gap: Actual GDP > potential GDP. The economy is "overheating" — operating above its sustainable capacity. Resources are being overworked, inflationary pressures build, and wages rise as firms compete for scarce labour.

Negative output gap: Actual GDP < potential GDP. There is spare capacity — unemployed workers and idle factories. The economy could produce more without causing inflation.

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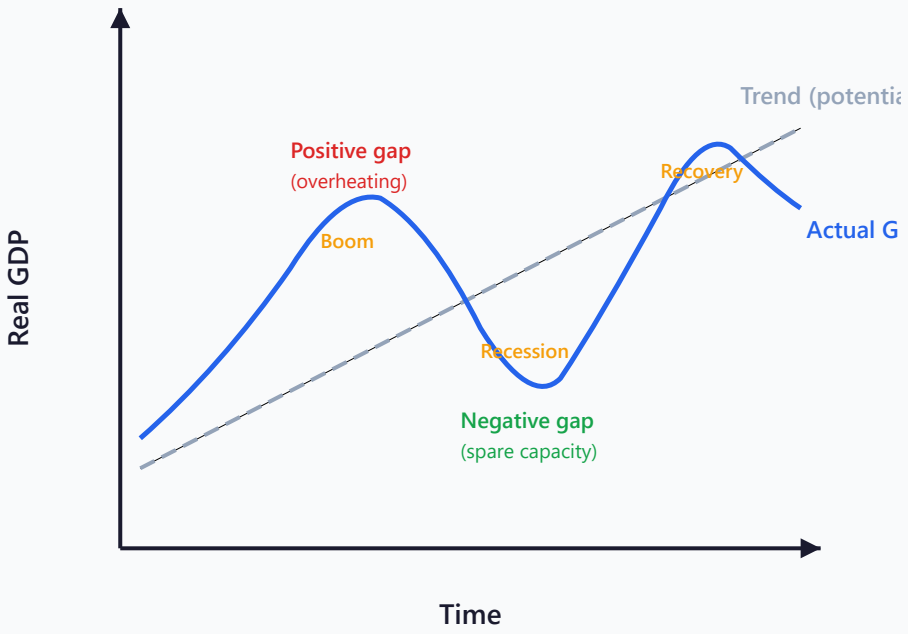

Figure 1: The economic cycle showing positive and negative output gaps relative to trend GDP

Implications of the Output Gap

	Positive Output Gap	Negative Output Gap
Employment	Unemployment below the natural rate; labour shortages in some sectors	Unemployment above the natural rate; spare labour capacity
Inflation	Rising (demand-pull inflationary pressure)	Falling or low (deflationary pressure)
Wages	Rising (firms compete for scarce workers)	Stagnant or falling
Policy response	Contractionary policy (raise rates, cut spending) to cool the economy	Expansionary policy (cut rates, increase spending) to stimulate demand

Inflation: Causes in Detail

Demand-Pull Inflation

Demand-pull inflation occurs when aggregate demand grows faster than aggregate supply, pulling prices up. It happens when the economy is at or near full capacity — "too much money chasing too few goods."

Causes of excess demand:

- Excessive growth in consumer credit and spending
- Large expansionary fiscal policy (tax cuts, spending increases)
- Excessively loose monetary policy (interest rates too low for too long)
- Rapid growth in the money supply (e.g., through QE)
- Depreciation of the exchange rate (boosting export demand and raising import costs)
- Strong global demand for UK exports

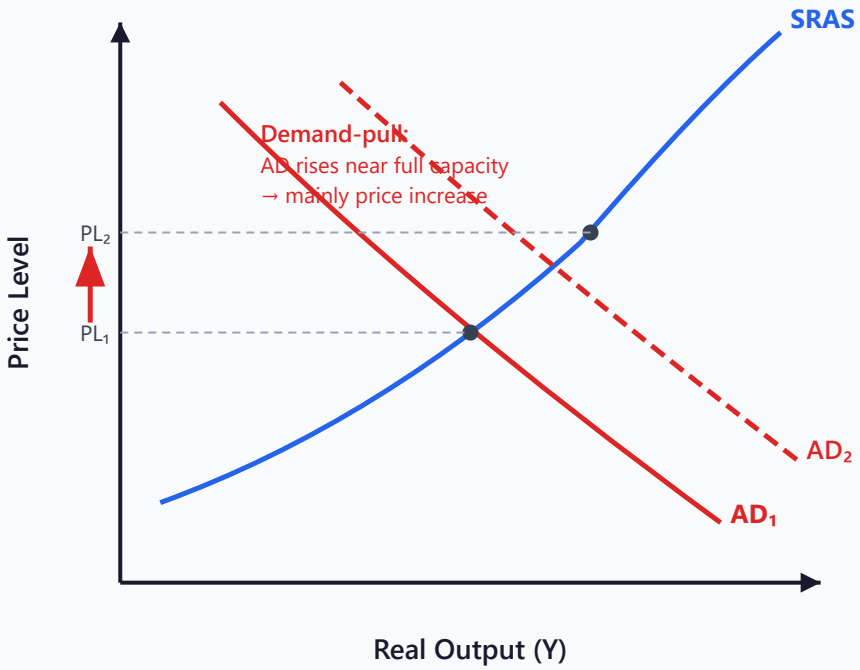

Figure 2: Demand-pull inflation — AD shifts right when the economy is near capacity, causing mainly a rise in the price level

Cost-Push Inflation

Cost-push inflation occurs when rising production costs push up prices across the economy. SRAS shifts left — firms pass higher costs onto consumers. The result is simultaneously rising prices AND falling output — known as **stagflation**.

Causes of rising costs:

- **Oil/energy price shocks:** Energy is an input in nearly every production process. A spike in oil or gas prices raises costs across the entire economy.
- **Rising wages above productivity growth:** If workers receive pay rises not matched by productivity gains, unit labour costs rise.
- **Exchange rate depreciation:** A weaker pound makes imported raw materials, components, and finished goods more expensive.
- **Rising commodity prices:** Global increases in the prices of food, metals, or other raw materials.
- **Higher indirect taxes:** An increase in VAT or excise duties directly raises the price level.
- **Supply chain disruptions:** Pandemics, wars, natural disasters that disrupt production and distribution.

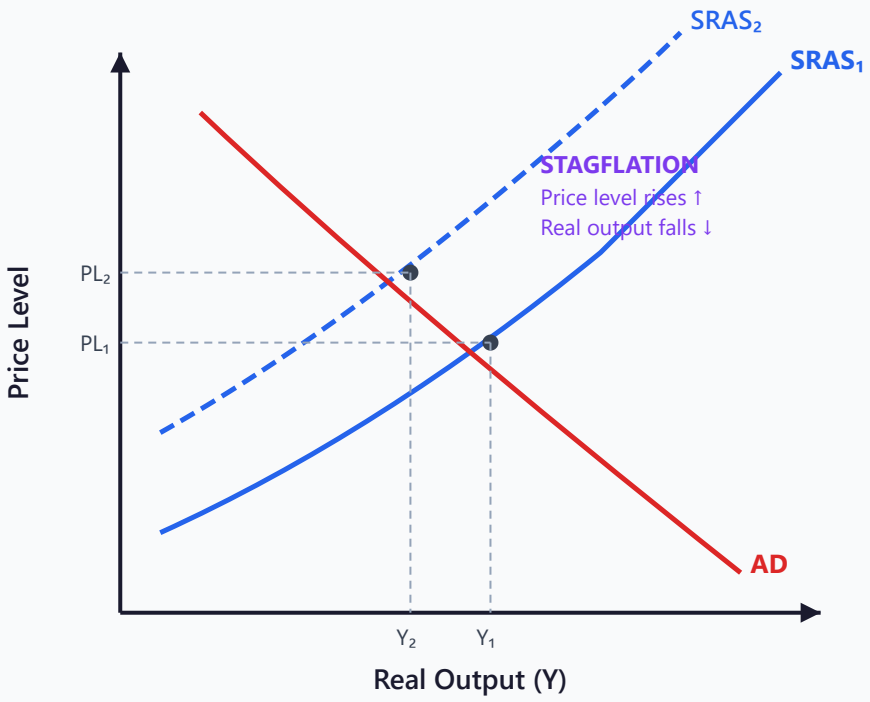

Figure 3: Cost-push inflation — SRAS shifts left, causing the price level to rise while output falls (stagflation)

Real-World Example: UK Stagflation 2022

The UK experienced textbook cost-push inflation / stagflation in 2022. The causes were supply-side: Russia's invasion of Ukraine spiked natural gas prices (400%+), global supply chains were still disrupted from COVID, and Brexit had increased trade friction with the EU. CPI inflation hit 11.1% in October 2022 while real GDP growth stagnated. The Bank of England faced a dilemma: raising interest rates to fight inflation would further slow growth and increase unemployment. This is the classic policy challenge of cost-push inflation — there is no painless solution.

Consequences of Inflation

Consequence	Explanation	Who Is Affected
Reduced purchasing power	Each pound buys fewer goods and services. Real incomes fall if wages don't keep pace.	Especially harmful for those on **fixed incomes** (pensioners, benefit recipients) whose income doesn't automatically adjust
Menu costs	Firms must frequently update price lists, catalogues, vending machines, and websites — a real cost of time and resources.	Firms, especially small businesses
Shoe-leather costs	People spend more time and effort shopping around for the best deals, moving money between accounts, and managing cash — "wearing out shoe leather."	Consumers

Reduced competitiveness	If UK inflation is higher than trading partners, UK exports become relatively expensive, reducing demand for UK goods abroad.	Export-oriented firms, current account
Uncertainty	Volatile inflation makes it harder for firms to plan investment and for consumers to make long-term financial decisions. This reduces economic efficiency.	Firms considering investment, consumers planning retirement
Fiscal drag	If income tax thresholds are not uprated with inflation, people are pushed into higher tax brackets even though their real income hasn't increased.	Workers — especially in the UK where thresholds were frozen 2021-28
Redistribution	**Borrowers gain** (the real value of their debt falls). **Savers lose** (the real value of their savings is eroded, especially if interest rates don't keep pace with inflation).	Mortgage holders benefit; cash savers lose
Wage-price spiral	Workers demand higher wages to compensate for inflation → firms pass higher wage costs on as higher prices → workers demand even higher wages → and so on.	The entire economy — inflation becomes self-perpetuating

Consequences of Deflation

Deflation (sustained falling prices) may seem beneficial to consumers, but it can be extremely damaging:

- **Delayed spending:** If consumers expect prices to fall further, they postpone purchases — reducing AD and deepening the downturn.
- **Rising real debt burden:** The real value of debt increases, making it harder for borrowers (households, firms, governments) to repay. This can trigger defaults and banking crises.

- **Falling asset prices:** House prices and share prices fall, creating a negative wealth effect and reducing consumer confidence.
- **Deflationary spiral:** Falling prices → lower profits → job cuts → less spending → prices fall further → a self-reinforcing downward spiral that is extremely difficult to break.
- **Monetary policy impotence:** Nominal interest rates cannot go below zero (the zero lower bound / liquidity trap). If inflation is negative, real interest rates are positive even at 0% nominal rates — the central bank cannot stimulate the economy effectively.

Real-World Example: Japan's Lost Decades (1990s-2010s)

Japan experienced deflation or near-zero inflation for much of the 1990s-2010s after a massive asset bubble burst in 1991. Property and share prices collapsed, banks accumulated bad debts, consumers saved rather than spent (expecting lower future prices), and businesses refused to invest. Despite interest rates being cut to 0% and massive government spending (Japan's national debt rose to over 250% of GDP), deflation persisted. Japan's experience is the cautionary tale for why central banks fear deflation — once it takes hold, it is extraordinarily difficult to escape.

The Phillips Curve

The Phillips Curve (A.W. Phillips, 1958) shows the empirical relationship between the rate of unemployment and the rate of inflation/wage growth. The original observation was an **inverse relationship**: when unemployment falls, inflation tends to rise, and vice versa. This suggests a short-run trade-off between the two objectives.

The Short-Run Phillips Curve (SRPC)

The SRPC shows the trade-off between inflation and unemployment for a **given level of inflation expectations**. Policymakers can move along the curve by adjusting aggregate demand:

- **Expansionary policy** (cut rates, increase spending) → moves the economy up the SRPC → lower unemployment but higher inflation
- **Contractionary policy** (raise rates, cut spending) → moves the economy down the SRPC → lower inflation but higher unemployment

Why the Trade-Off Exists

When unemployment falls (the economy is growing):

- Labour markets tighten → firms must offer higher wages to attract workers → wage-push inflation
- Consumer spending rises → demand-pull inflationary pressure
- Firms have greater pricing power as demand is strong

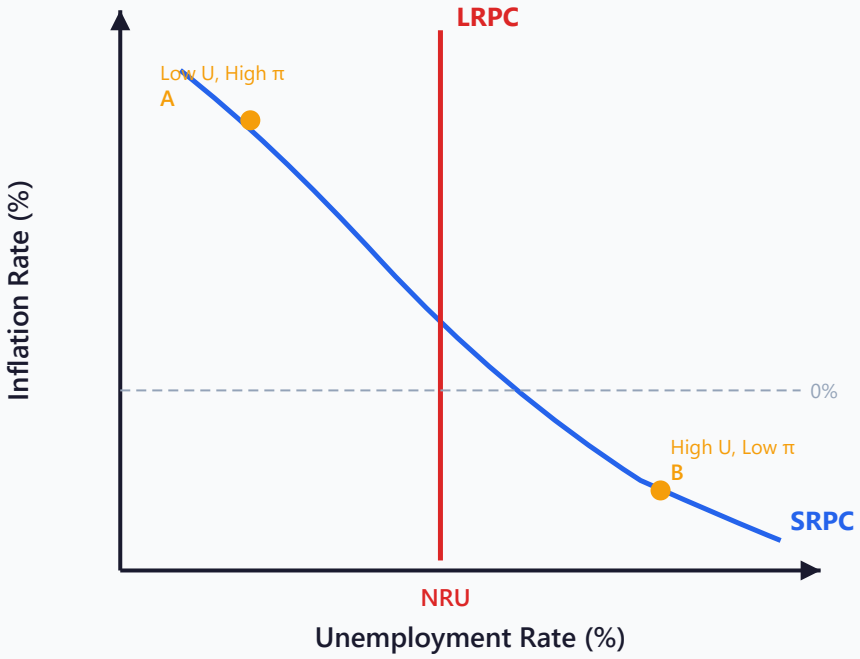

Figure 4: The Short-Run Phillips Curve (SRPC) shows the trade-off. The Long-Run Phillips Curve (LRPC) is vertical at the Natural Rate of Unemployment (NRU).

The Long-Run Phillips Curve (LRPC) — The Monetarist/Classical View

Milton Friedman and Edmund Phelps argued (1968) that the Phillips Curve trade-off is **only short-run**. In the long run, the Phillips Curve is **vertical** at the Natural Rate of Unemployment (NRU).

The argument:

1. If the government boosts AD to reduce unemployment below the NRU, inflation rises (moving up the SRPC).

2. Workers initially accept the higher wages in nominal terms, not realising prices have risen too (money illusion).
3. Eventually, workers adjust their expectations: they realise inflation has eroded their real wage, and demand even higher nominal wages.
4. This pushes up costs, firms lay off workers, and unemployment returns to the NRU — but now at a **higher rate of inflation**.
5. The SRPC has shifted upward. Any attempt to reduce unemployment below the NRU permanently will only result in **accelerating inflation**.

> **The Natural Rate of Unemployment (NRU)** is the rate of unemployment that exists when the labour market is in equilibrium. It consists of **frictional and structural unemployment** — not cyclical. It is the rate consistent with stable inflation, also called **NAIRU** (Non-Accelerating Inflation Rate of Unemployment). Estimated at about 4-5% for the UK.

Reducing the NRU

The NRU can only be reduced through **supply-side policies** that shift the LRPC to the left:

- **Education and training:** Reduce structural unemployment by equipping workers with relevant skills
- **Labour market reform:** Improve job-matching through better information, employment agencies, and flexible contracts
- **Welfare reform:** Reduce benefit dependency and increase incentives to seek work
- **Reducing trade union power:** Lower wage rigidity (Classical/Monetarist view)
- **Geographical mobility:** Housing policies that make it easier for workers to move to where jobs are

> **Real-World Example: The Phillips Curve Breakdown?**
> The Phillips Curve relationship has been questioned in recent decades. In the 2010s, many countries (UK, US, Japan) experienced very low unemployment but inflation also remained low — seemingly contradicting the curve. Possible explanations: globalisation (cheap imports keeping prices down), weaker trade unions (less wage bargaining power), gig economy (suppressing wage growth even at low unemployment), and well-anchored inflation expectations (central banks' credibility keeping expectations at 2%). However, in 2021-22, as unemployment fell sharply after COVID, inflation surged — suggesting the trade-off may reassert itself when labour markets are truly tight.

Policy Conflicts and Trade-Offs

Governments cannot achieve all macroeconomic objectives simultaneously. Pursuing one objective often comes at the cost of another:

Conflict	Explanation	Example
Growth vs Inflation	Rapid demand-led growth can cause demand-pull inflation when the economy approaches full capacity.	The UK's Barber Boom (1972-73) — aggressive fiscal expansion caused inflation to hit 25%
Low Unemployment vs Inflation	The Phillips Curve trade-off — reducing unemployment below the NRU creates inflationary pressure.	Bank of England had to raise rates in 2022-23 despite risking higher unemployment
Growth vs Current Account	Economic growth increases consumer incomes → higher spending on imports → current account worsens.	UK's persistent current account deficit widens during periods of strong domestic growth
Growth vs Environment	Higher production typically means more carbon emissions, pollution, and resource use.	China's rapid growth caused severe air and water pollution
Low Inflation vs Unemployment	Tight monetary policy (high rates) controls inflation but slows growth and may increase unemployment.	Volcker's rate hikes in the US (1979-82) crushed inflation but caused a severe recession
Growth vs Equality	Growth may disproportionately benefit capital owners and high-skilled workers, increasing inequality.	UK growth 2010-2019 saw wages stagnate for the bottom 50% while asset prices soared

Exam Tip: When discussing any macroeconomic policy, always analyse the potential **conflicts with other objectives**. The best exam answers explain: (1) how the policy achieves its primary objective, (2) which other objectives it might compromise, (3) whether the trade-off is worth making given the

current economic context, and (4) whether there are complementary policies that could mitigate the conflict.

Evaluation: Growth — At What Cost?

Key Evaluation Framework for Growth Questions:

- **Quality of growth matters:** Is growth driven by sustainable investment and productivity improvement, or by unsustainable debt-fuelled consumption?
- **Distribution matters:** Who benefits? Growth that only enriches the top 10% while leaving the bottom 50% behind is less valuable than inclusive growth.
- **Environmental sustainability:** Growth that destroys natural capital (forests, clean air, biodiversity) may reduce future welfare even as it raises current GDP.
- **Short-run vs long-run:** Demand-led growth is quick but potentially inflationary and temporary. Supply-side growth is slow but sustainable and non-inflationary.
- **Composition matters:** Growth driven by government military spending is very different from growth driven by business investment in technology.
- **GDP is an incomplete measure:** Growth in GDP doesn't necessarily mean growth in wellbeing, happiness, or quality of life. Alternative measures (HDI, GPI) tell a fuller story.

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